

CASE STUDY: AUTOMATED SOFTWARE ENGINEERING

The 45-Minute CRM

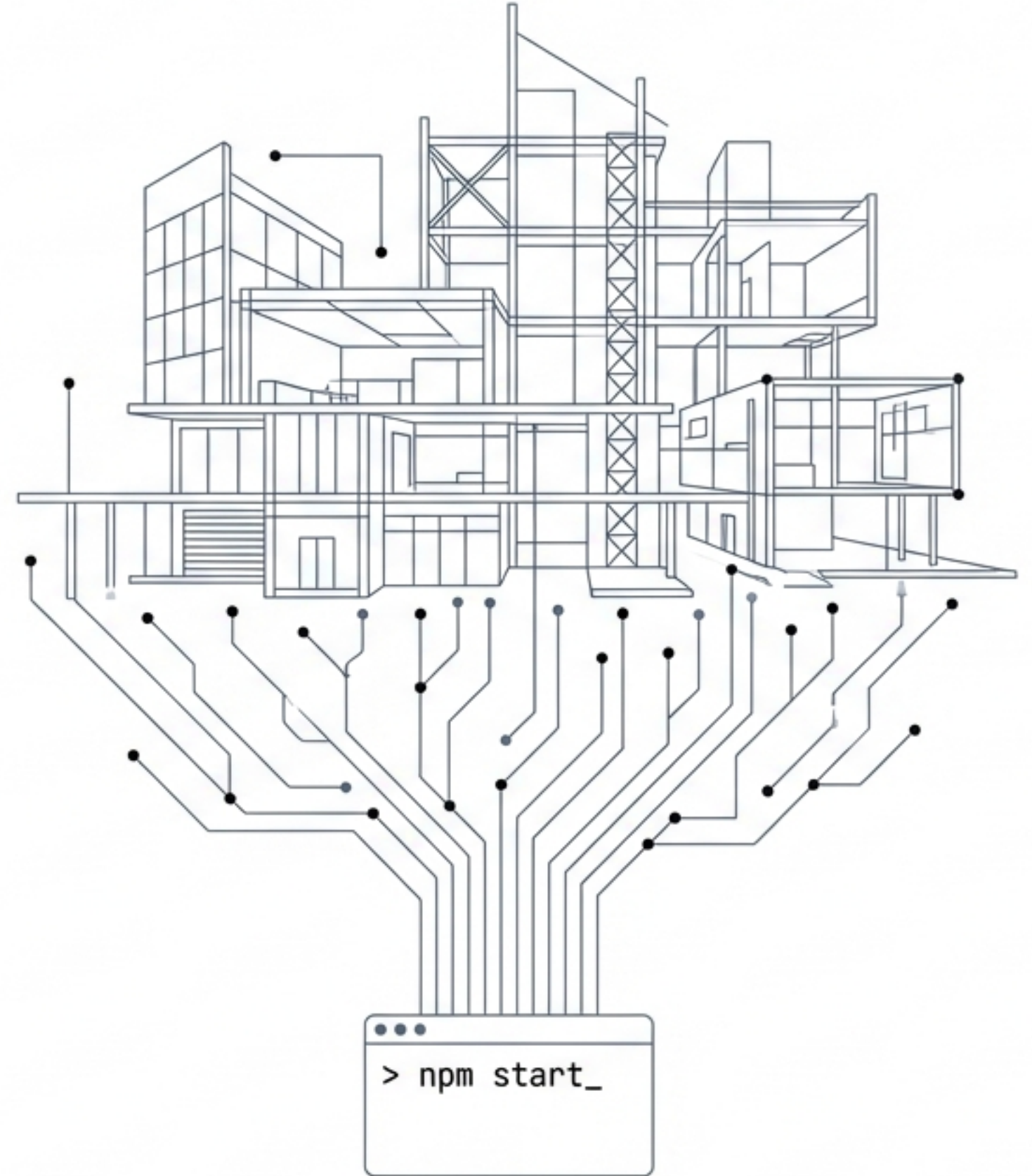
Building a production-ready system with Claude-Flow & Hybrid Agent Architecture.

PROJECT:
CRM Pro

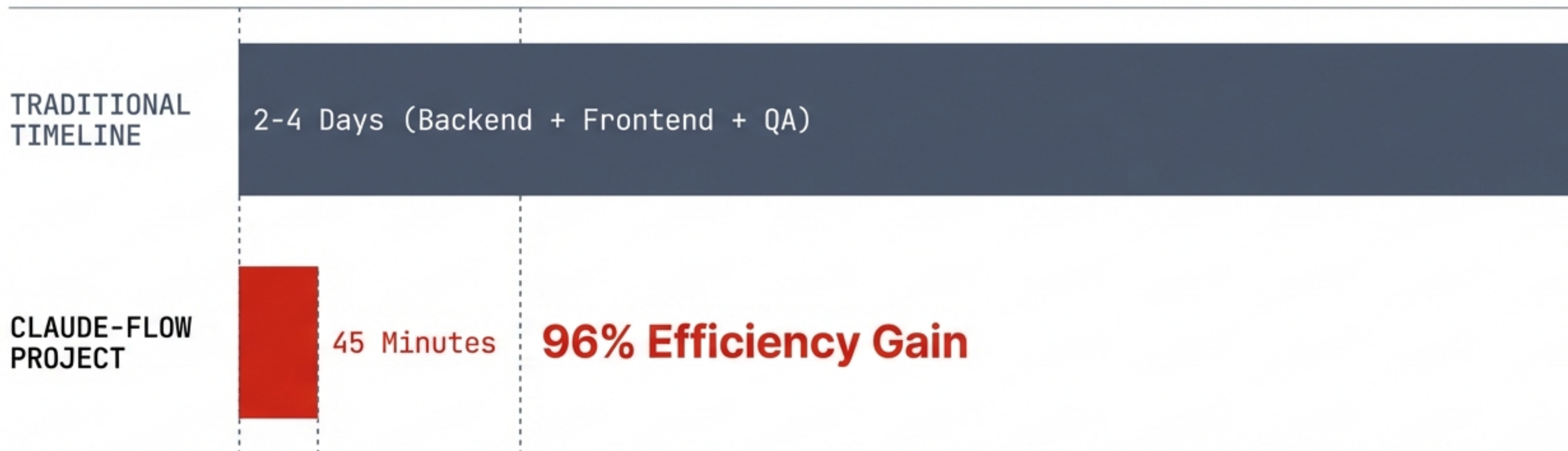
DATE:
26 January 2026

METHODOLOGY:
Hybrid (Hive-Mind + Direct)

DURATION:
45 Minutes



Software estimation usually deals in days or weeks. We are proposing program mines.

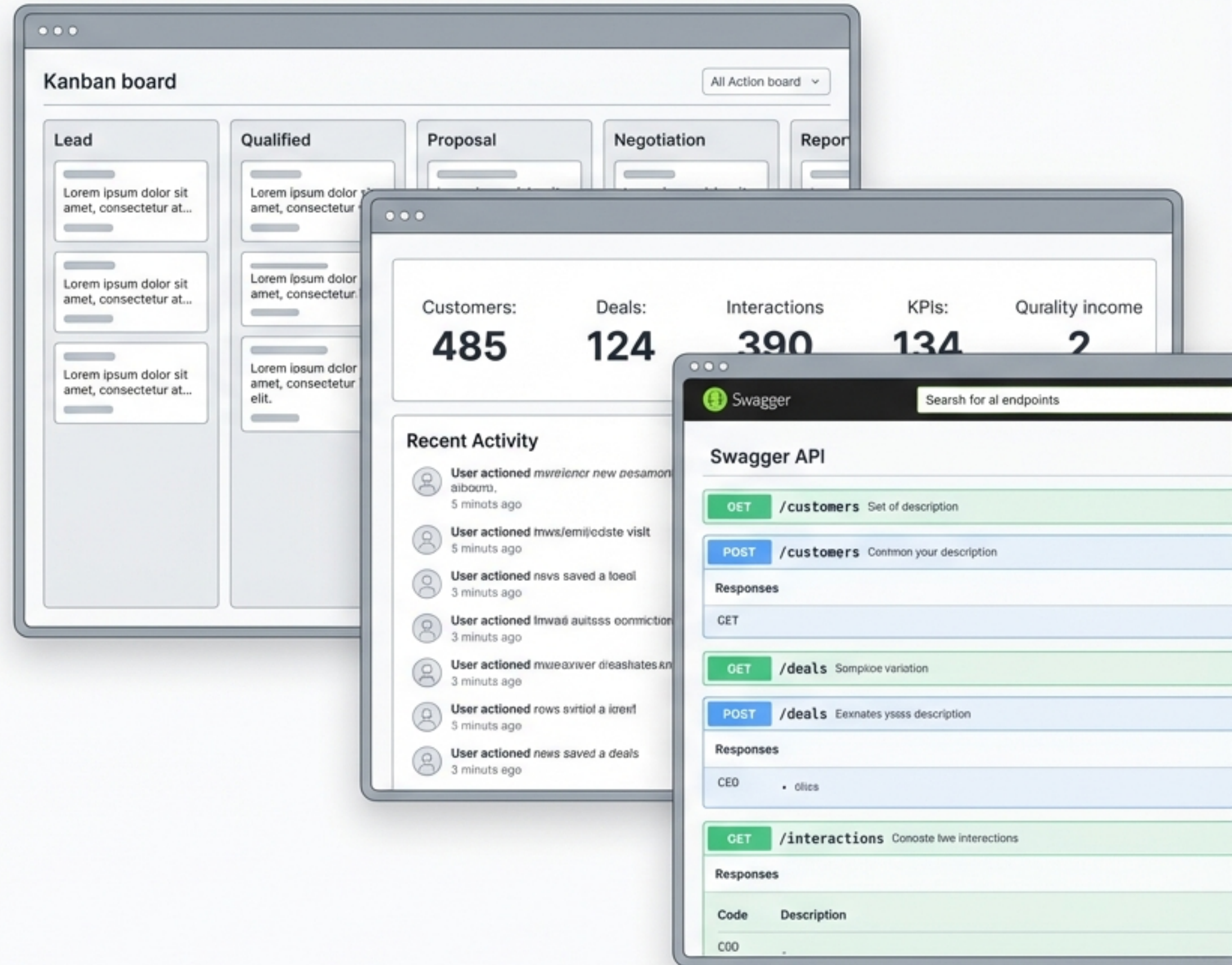


From multi-day cycles to sub-hour delivery.

Speed means nothing without utility. We delivered a fully functional system.

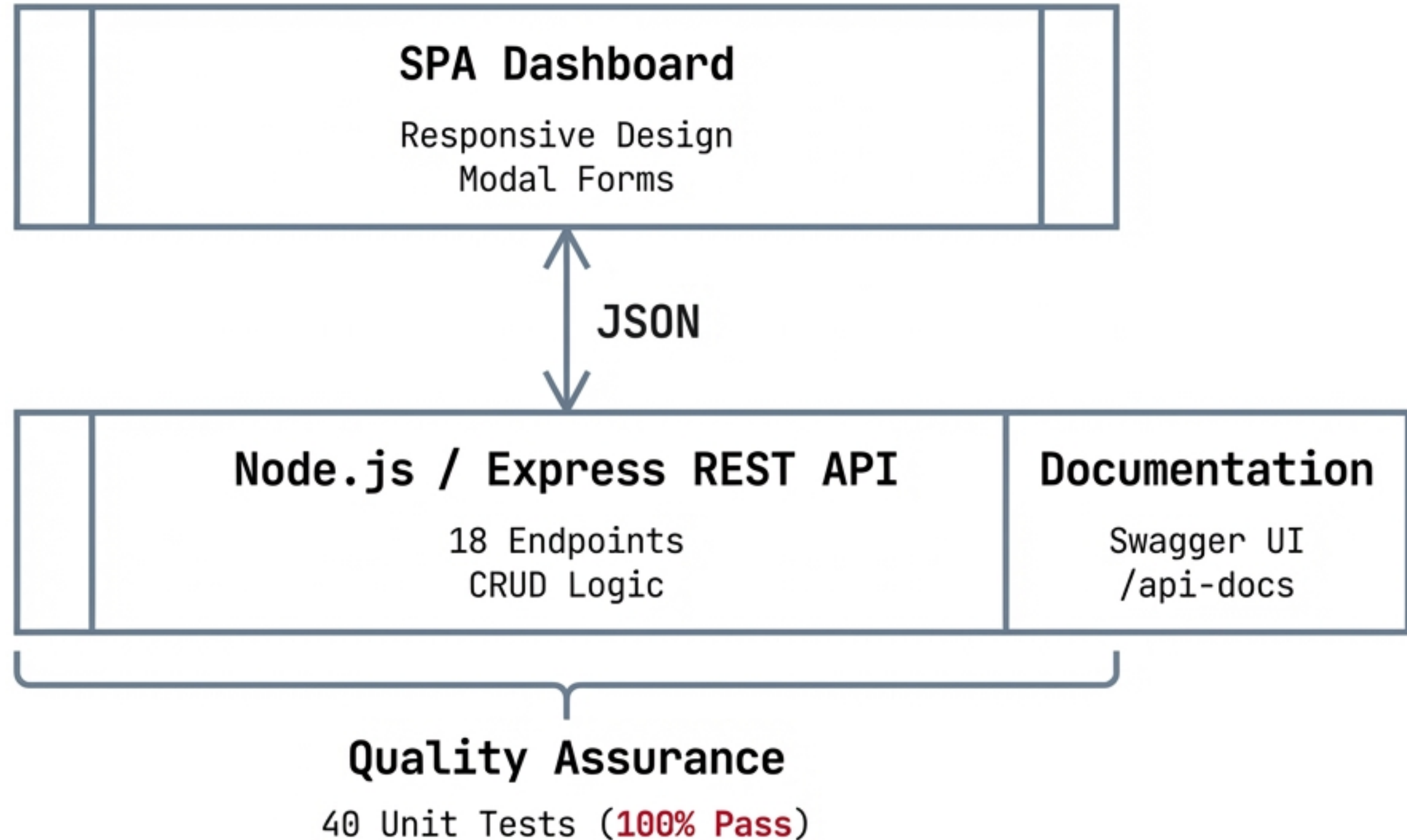
// Delivered Items

- [✓] REST API: 18 endpoints
- [✓] Deals Pipeline: 6 Stages
- [✓] Interactions: 4 Types
- [✓] Customer Management: CRUD
- [✓] Docs: OpenAPI 3.0 Specs



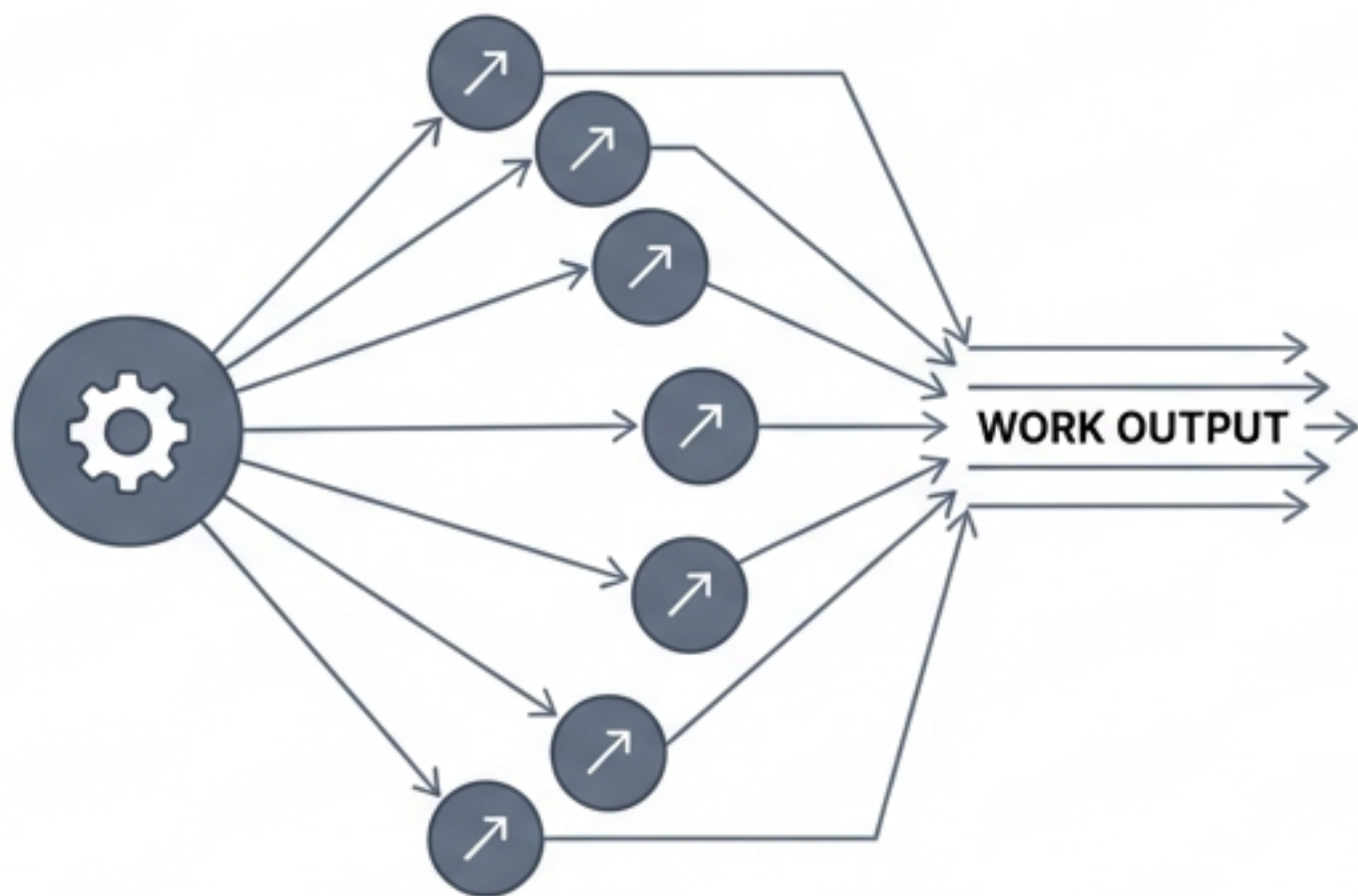
A robust, modern stack built by intelligent automation.

Clean separation of concerns, pre-seeded with realistic data.



The Hybrid Advantage: Swarm Intelligence Meets Surgical Precision

MODE 1: SWARM ORCHESTRATION



Function: 8 agents in parallel

Use Case: Large Features, Architecture

Benefit: 5x Speedup

MODE 2: DIRECT ASSISTANCE

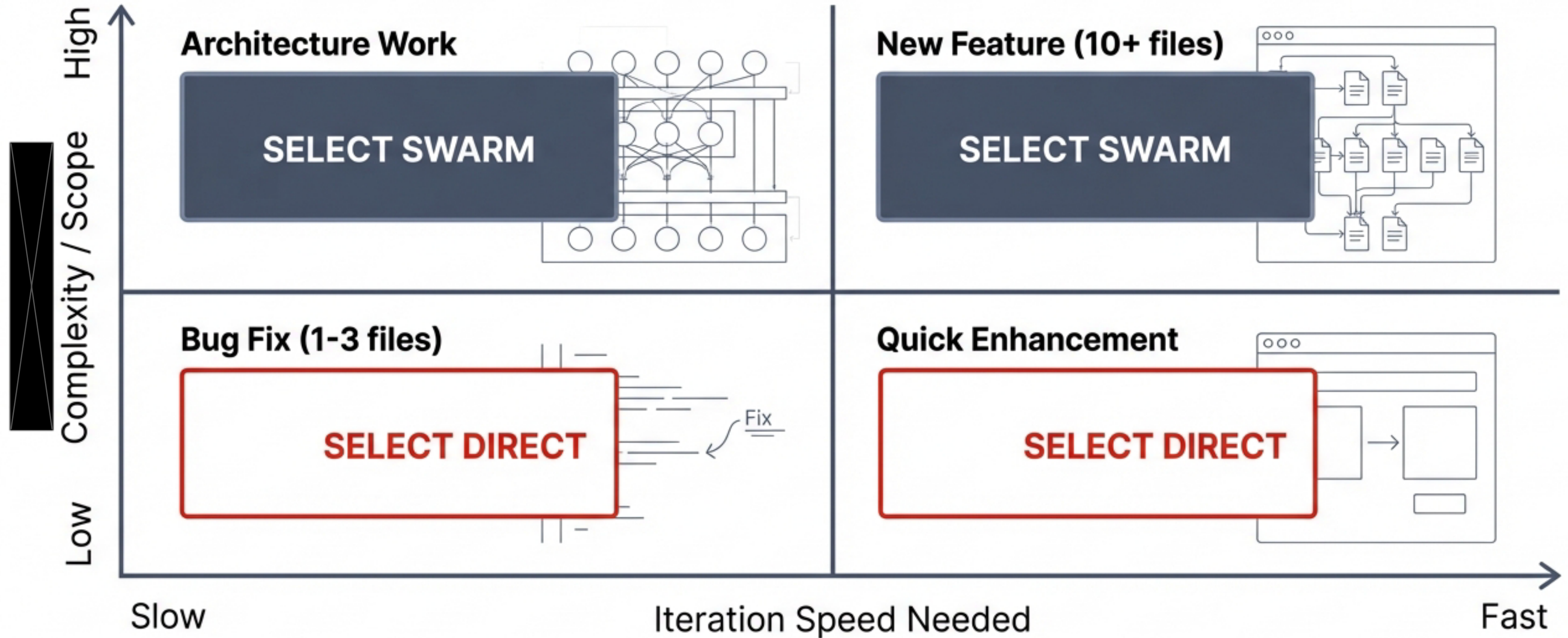


Function: Single Focus Agent

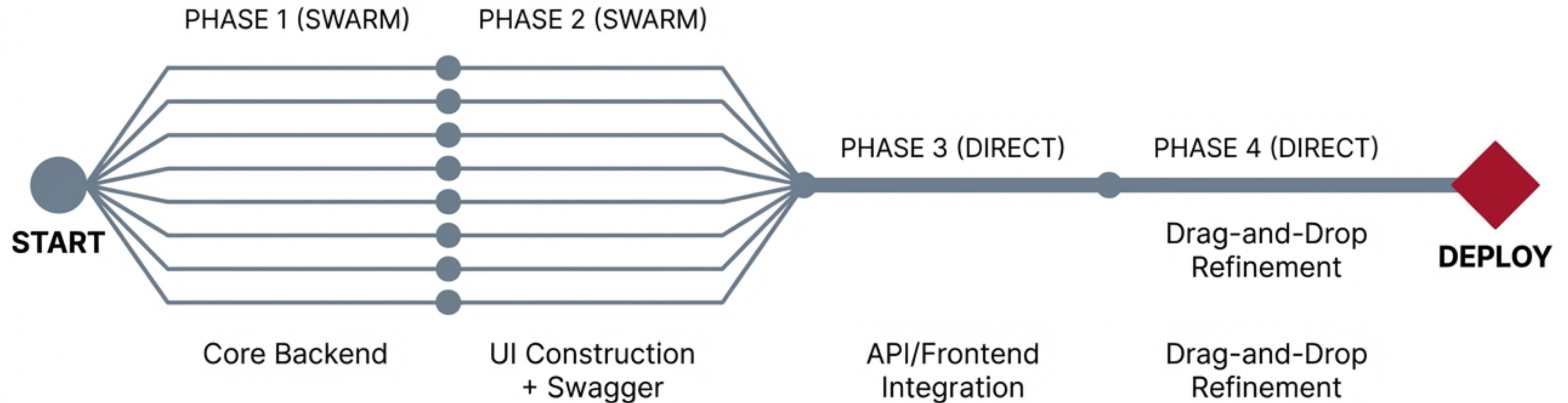
Use Case: Bug fixes, Polish

Benefit: Immediate Iteration

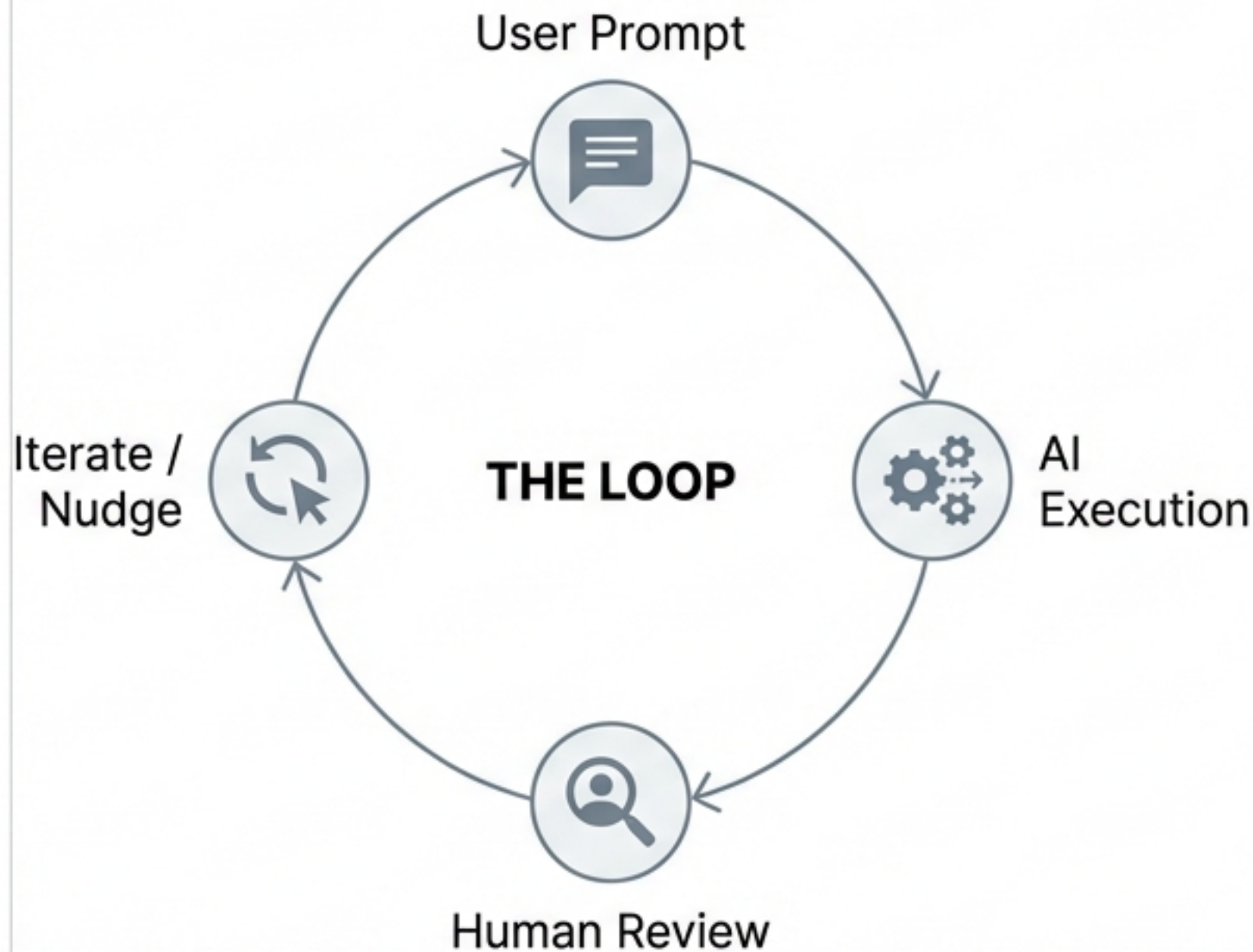
The Decision Matrix: Selecting the right agent for the task.



From "One Command" to Deployment in four phases.



Human-in-the-Loop: AI provides the horsepower; humans provide the steering.



The 'Nudge' Moments



Can you check if the storage agent is stuck?



Resolved: Notification was lost, but task was complete.



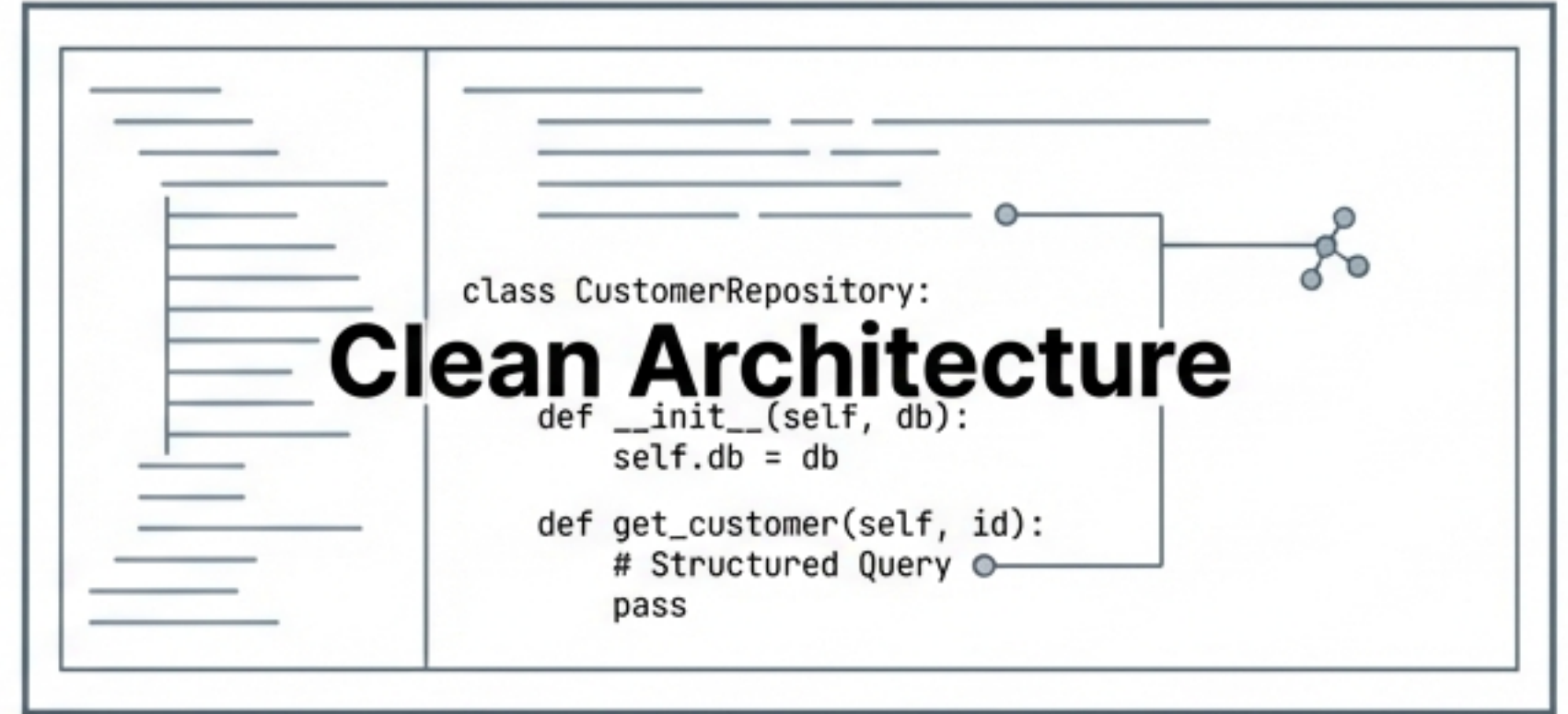
Can you check if it is not stuck?



Confirmed: Backend files created successfully.

Speed did not compromise stability or documentation.

```
PASS ./tests/customer.test.js ✓  
PASS ./tests/deals.test.js ✓  
  
Test Suites: 40 passed, 40 total
```



100%

Test Pass Rate



OpenAPI 3.0 Specs



Responsive UI

The Business Case: 96% Efficiency Gain

Phase	Traditional Time	Claude-Flow Time	Improvement
Backend Dev	1-2 Days	10 Minutes	95%+
Frontend Dev	1-2 Days	15 Minutes	95%+
Bug Fixes/Polish	4-8 Hours	20 Minutes	90%+

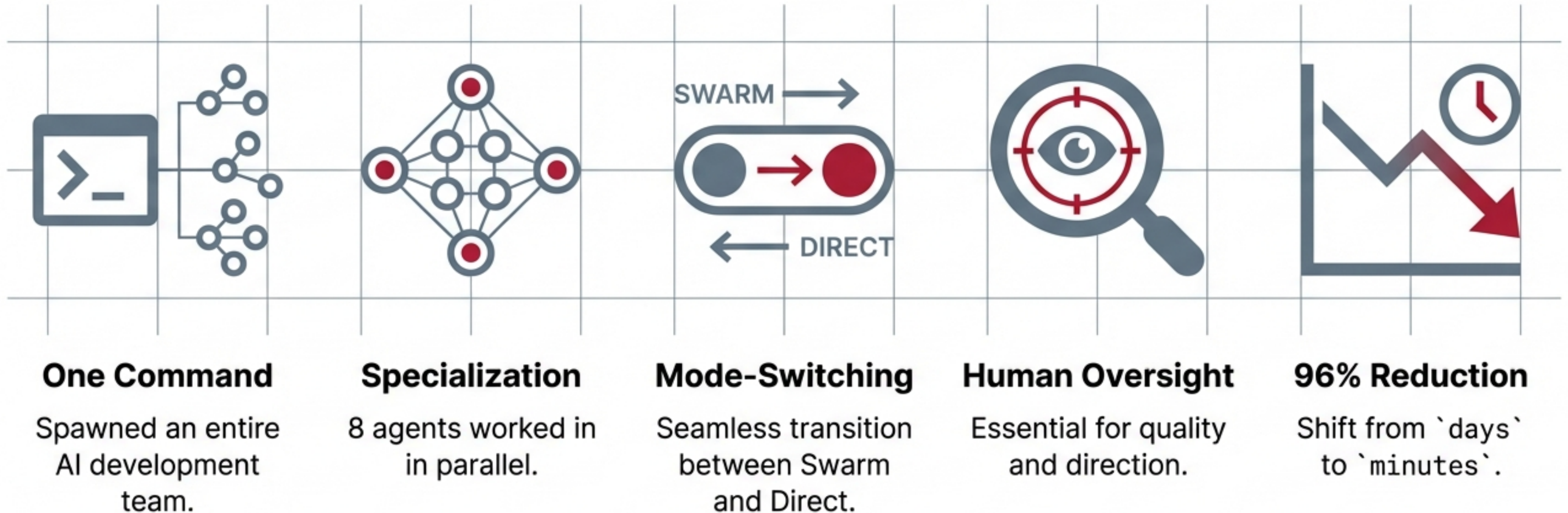


Significant time reduction
achieved through AI integration.



TOTAL: 2-4 DAYS → 45 MINUTES

The future of software development is Hybrid AI Orchestration.



References & Resources

Project Documents

- 📁 > 001-crm-development-debrief.md
- └─ > 002-conversation-flow-analysis.md
- └─ > 003-architecture-by-phase.md
- └─ > 004-user-stories-and-features.md
- └─ > 005-claude-flow-vs-claude-code.md

External Resources

- 🔗 Claude Flow: [A Beginner's Guide](#) by [Vatsal Shah](#)
- 🔗 GitHub Repository: [Claude Flow](#)

Project Location: /home/claude/workspace

API Docs: <http://localhost:3000/api-docs>